THALITWIN TECHNICAL ARCHITECTURE BLUEPRINT

Happiest Health Digital Twin Challenge 2026 · Stepped-Care Frugal Engine

1. INGESTION & QC

- PhysioNet CGMacros (45 Adults)
- Dexcom G6 Pro (5m) & Libre Pro
- **Sensor Noise Floor (38.7 mg/dL)**
- 5-min Resampling & Gap Impute

2. SENSOR EMULATOR

- T1 Rich: 5-min Continuous CGM
- **T2 Frugal: Sparse Glucometer**
  (7% ISO-15197 Meter Noise)
- T3 Triage: Fitbit Wearable Only

3. THALI LIBRARY

- ICMR-NIN IFCT 2017 Registry
- 25 Authentic Indian Dishes
- Net Carbs, Protein, Fat, Fiber
- Glycemic Load & Index Maps

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4. LEAKAGE-FREE FEATURE STORE & GROUPKFOLD PARTITIONING

- Strictly Past-Only Features: Slopes (15m, 30m), Rolling Glycemic Variability (CV, SD), Minutes Since Last Reading
- Stratified GroupKFold (5-Splits by Subject ID): Zero participant overlap between training and held-out validation folds
- Leakage Guard: Excursion windows included ONLY when baseline glucose < 180 mg/dL (proved by tests/test_leakage.py)

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5. TWIN CORE: EMPIRICAL-BAYES

- Population EHR Prior: Age, BMI, HbA1c
- Shrinkage: w = n / (n + 4.0)
- **Learning Curve: Day 0 (44 mg/dL) -> Day 7 (30 mg/dL)**

6. TRUST & UNCERTAINTY LAYER

- Split-Conformal 90% Prediction Intervals
- Alert Budget: <= 1.0 False Alert / Day
- Monotonicity & Domain Abstention Guards

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7. SERVING & INTEROPERABILITY: Doctor Console (Streamlit + Plotly) · Replay Clock · ABDM FHIR Exchange

Production Path: Device MQTT -> InfluxDB Time-Series -> FastAPI Inference Worker -> ABDM / Ayushman Bharat Digital Exchange